



Progress Report

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Respiratory Inductance Plethysmography (RIP) Device Prototype Development

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Goal

- To develop a Respiratory Inductance Plethysmography (RIP) device from components.
- To collect and analyze plethysmography data of lung function tests (LFT) including forced vital capacity (FVC), forced expiratory volume in first second (FEV₁) and FEV₁/FVC ratio and use it for diagnostic purposes through implementing machine learning methods [1].
- Primary non-specific diagnosis could be obstructive or restrictive type lung disease.
- Further specific diagnosis could be selective identification of a specific lung disease (e.g. COPD, pneumonia).

Components of the proposed RIP prototype device

- PWM DC current source exciter, operating at fixed 50% duty cycle and high frequency range (estimated: 100-500 kHz) [2].
- LC tank circuit having a resonant frequency f_s slightly below than the frequency of the exciter circuit.
- RIP belt as inductor in LC tank circuit, where resistance of the belt is considered as the real part of the impedance of the real inductor.
- Demodulator to filter out high frequency exciter signal, thus keeping low frequency signal obtained by the change in inductance of the belt due to expansion and contraction of the chest cavity.
- Analog signal preprocessing components of the signal if necessary (i.e. noise filter).
- Storing the data into secondary storage (i.e. MicroSD card in binary format) through a microcontroller based embedded system for further analysis.

Factors affecting sensitivity of the device

- Inductance of the belt (smaller inductance has higher sensitivity as seen in simulation).
- Resistance of the belt (optimum resistance values for sharp gain at f_s).
- Resonant frequency.

Works done

- A 30cm long and 2.5cm wide having a step size of 1cm, 24SWG enameled copper wire, sewed on elastic material, nearly sinusoid shaped belt has been constructed and tested for inductance measurements in various lengths [3].
- Simulation of the partial proposed circuit including LC tank circuit and a preamplifier has been run, observing frequency response curve and output signal and its changes due to change in inductance in Proteus.

Difficulties faced with hardware part

- Proper measurement of the inductance of the constructed belt could not be taken due to improper settings of the LCR meter used (GW Instek LCR819).

Proposed solution:

Ideal settings for inductance measurement should be:

- Series connection (for small inductance measurement).
 - Test frequency: 100kHz (the highest the LCR meter used can provide).
 - DC resistance of the belt also needs to be measured with proper settings for proper simulation and circuit design.
- Proper bending of the copper wire according to shape is challenging, as regularity of the shape is crucial for maintaining its inductance while stretching and relaxing back.

Proposed solution:

- Simple, hand-made wire bending tool could be constructed.
- Maintaining the shape of the copper wire while stretching and relaxing back is difficult.

Proposed solution:

- Stitching the whole copper wire with the elastic material could be effective in maintaining the shape, but stretchability of the belt might become affected.
- Stitching on peaks and middle points of the sinusoids.
- Off-the shelf low-cost disposable belts could be used if possible.

Data obtained from simulation

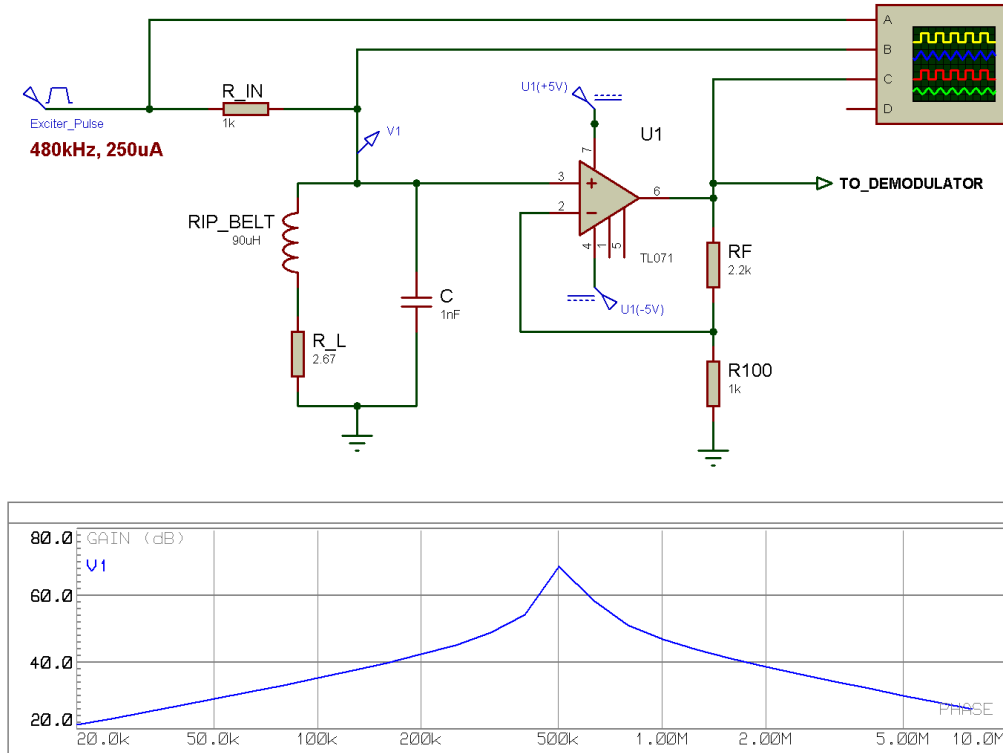


Figure 1: Frequency response for 90 μ H inductance of the belt.

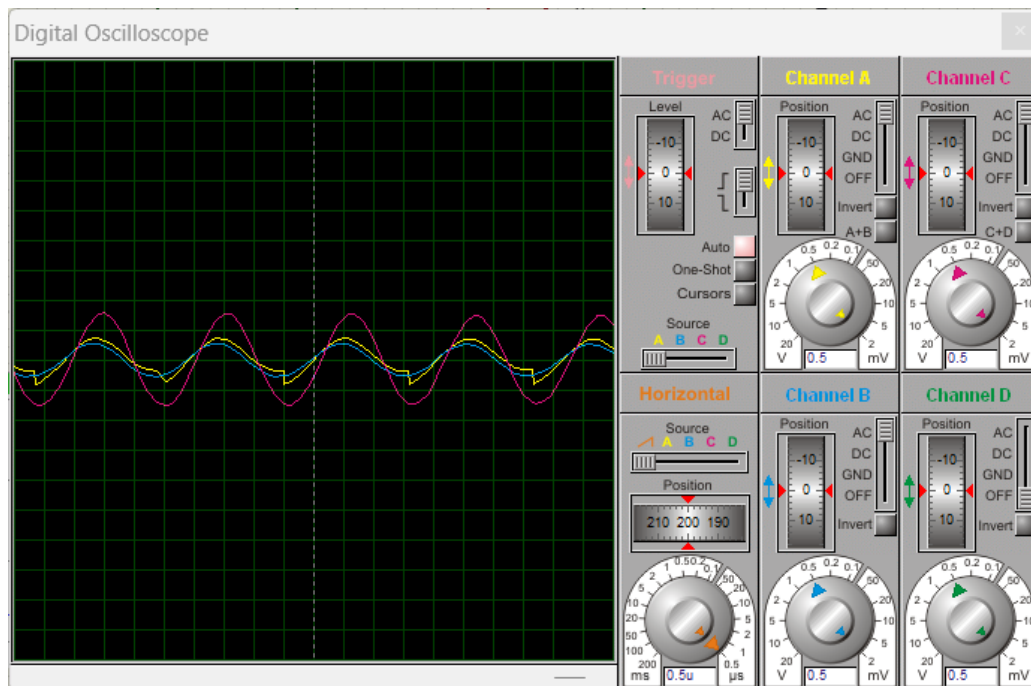


Figure 2: Output voltage (red) for 90 μ H.

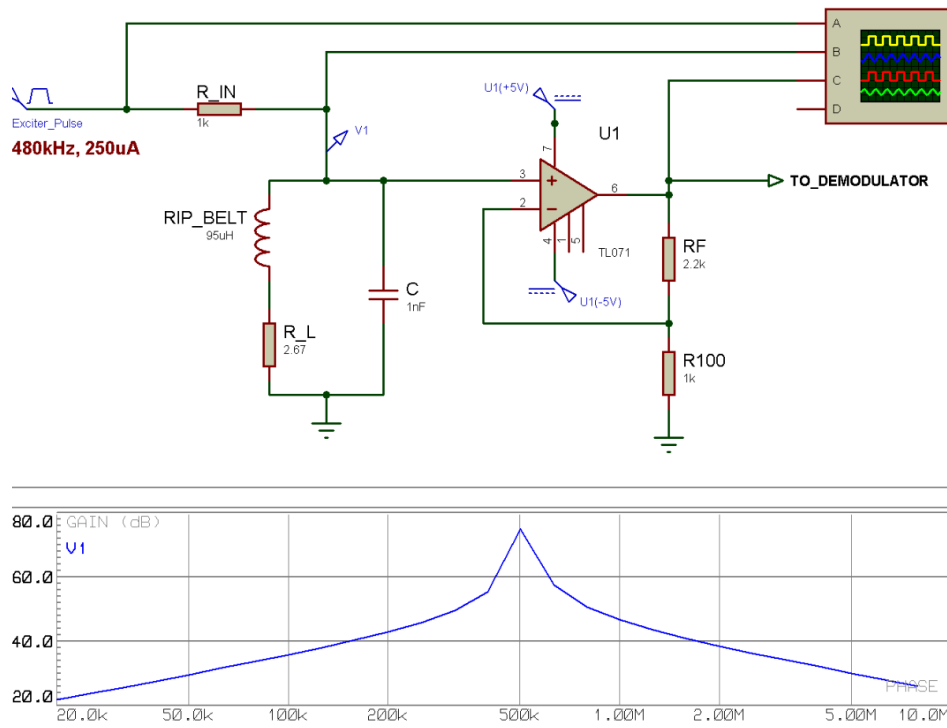


Figure 3: Frequency response for 95 μ H inductance of the belt.

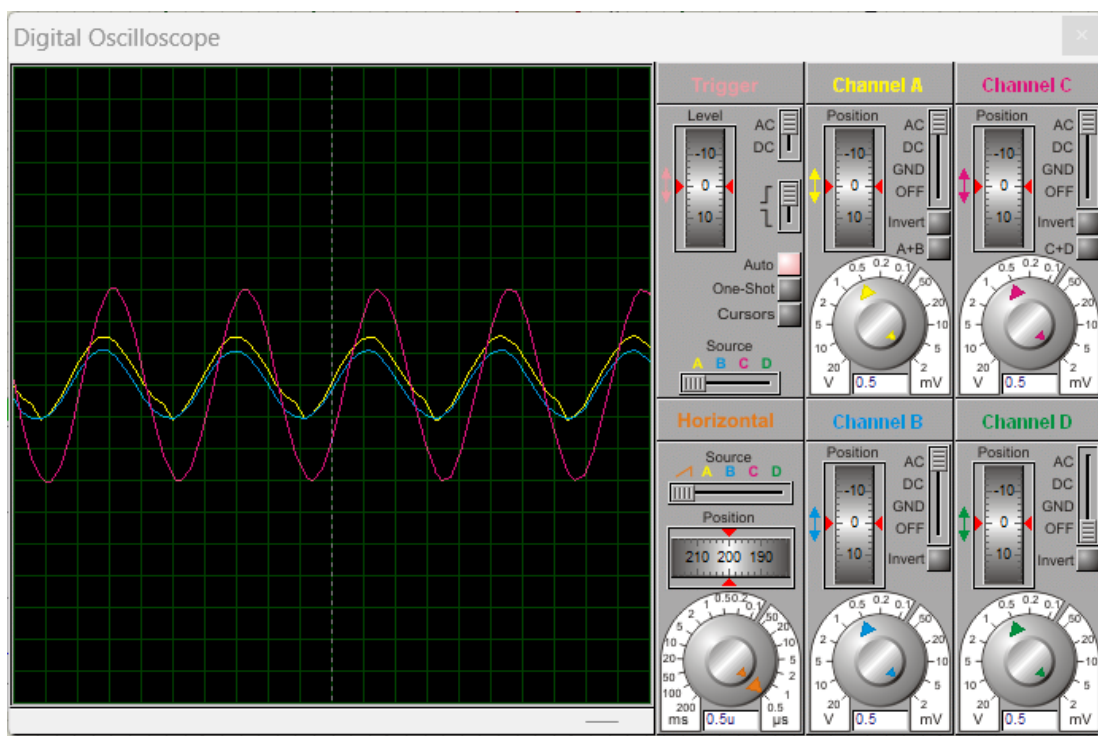


Figure 4: Output voltage (red) for 95 μ H.

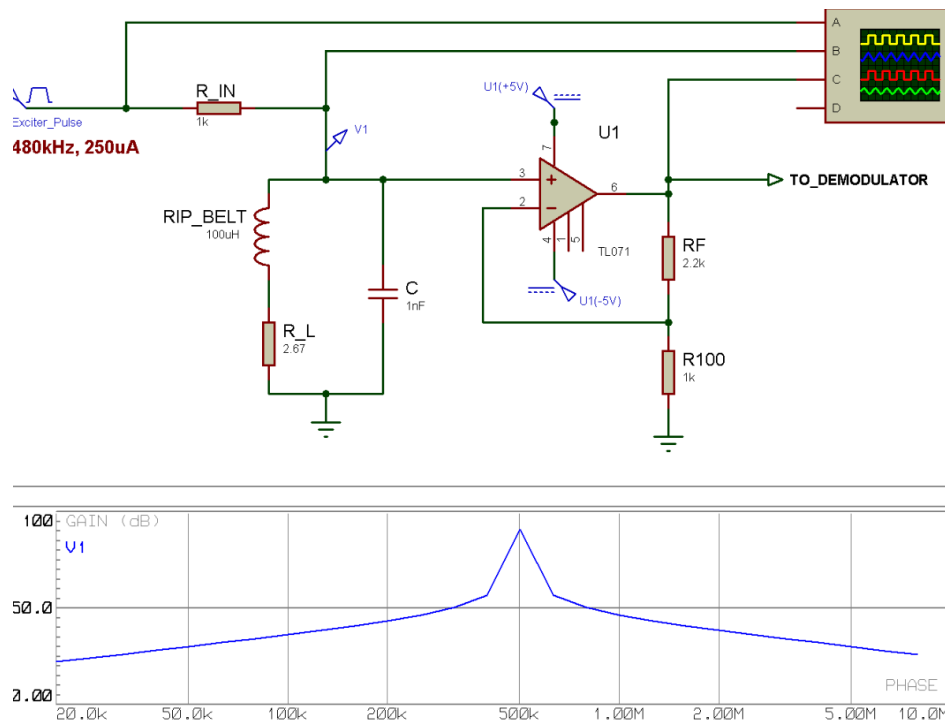


Figure 5: Frequency response for 100 μ H inductance of the belt.

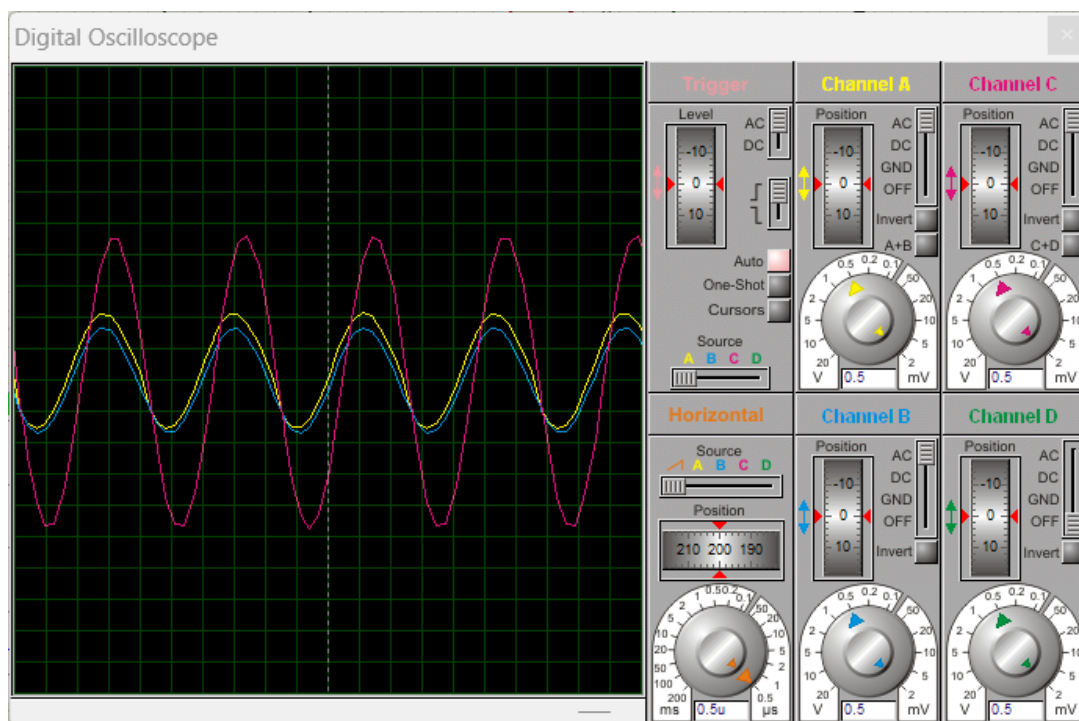


Figure 6: Output voltage (red) for 100 μ H.

Calculation of DC resistance of copper wire

Resistance of 24 SWG enameled copper wire: $0.0703 \, \Omega/\text{m}$ (approx.)

For 1cm step and 30cm long belt, wire length is calculated with the arc length formula:

$$l = \int_0^{2\pi} \sqrt{1 + \cos^2 x} \, dx \text{ where } f(x) = \sin x, \text{ thus } f'(x) = \cos x$$

Calculated wire resistance, $R_L = 30\text{cm} \times 0.01 \times 8.877 = 2.663 \, \Omega$

References

- [1] A. Laouani and S. Rouatbi, “Thorax and Abdomen Motion Analysis in Patients with Obstructive Diseases,” *J. Pulm. Respir. Med.*, vol. 06, no. 01, 2016, doi: 10.4172/2161-105X.1000313.
- [2] Z. Zhang, J. Zheng, H. Wu, W. Wang, B. Wang, and H. Liu, “Development of a Respiratory Inductive Plethysmography Module Supporting Multiple Sensors for Wearable Systems,” *Sensors*, vol. 12, no. 10, pp. 13167–13184, Sep. 2012, doi: 10.3390/s121013167.
- [3] H. T. Ngo, C. V. Nguyen, T. M. H. Nguyen, and T. Van Vo, “A Portable Respiratory Monitor Using Respiratory Inductive Plethysmography,” in *4th International Conference on Biomedical Engineering in Vietnam*, vol. 49, V. V. Toi, N. B. Toan, T. Q. Dang Khoa, and T. H. Lien Phuong, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2013, pp. 222–225. doi: 10.1007/978-3-642-32183-2_57.